



# Galaxy classification based on deep learning

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## Abstract

In this study, a deep learning-based galaxy classification method is discussed. The classification of galaxies is important for understanding the formation and evolution of the universe, and traditional classification methods rely on artificial visual analysis, but in the face of large amounts of data, this method is time-consuming and prone to error. In recent years, automated classification methods, especially using deep learning techniques, have gradually come into focus. Deep learning models, particularly convolutional neural networks (CNNs), are capable of automatically extracting and learning complex features in galactic images, enabling efficient and accurate classification. The research plan is to integrate multimodal data, train models with large-scale datasets, and introduce interpretative analysis into the classification process to improve model transparency. Ultimately, the goal is to develop an efficient galactic classification system to support data processing and analysis in the field of astronomy.

## CCS Concepts

• **Computing methodologies** → Artificial intelligence; Computer vision; Computer vision tasks; Visual inspection.

## Keywords

Deep learning, Image classification, Galaxy, Probability distribution

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## 1 Introduction

### 1.1 Research background and significance

The Galaxy is the basic celestial building block in the universe, made up of stars, planets, gas, dust and dark matter. Studying the morphology and classification of galaxies is important for understanding the formation and evolution of the universe. The traditional classification of galaxies relies on astronomers' visual analysis of galaxy images and usually classifies galaxies into several categories, such as elliptical, spiral and irregular galaxies, based on their morphological characteristics. However, this classification method is limited by the subjectivity and resolution of human vision, and it is difficult to process large amounts of galactic data. [1]

In recent years, with the advancement of astronomical observation technology, especially the wide use of large astronomical telescopes and astronomical satellites, the quantity and quality of galaxy images obtained have increased rapidly. This explosion of data brings great challenges to analysis and classification, and also provides possibilities for the application of automated classification methods. Deep learning, as an important branch of artificial intelligence, has shown strong capabilities in image recognition and classification, and has gradually become an important direction of galactic classification research.

### 1.2 Existing methods of classifying galaxies

The existing galaxy classification methods mainly include two categories: manual classification and automatic classification based on images. Manual classification is the earliest method, by astronomers based on the image's morphological characteristics, such as the brightness distribution of galaxies, spiral arm structure, etc., to classify galaxies. Hubble's galaxy classification system is a classic example of this approach and is still widely used today.

However, manual classification methods are time-consuming and subject to subjective factors. In order to solve these problems, automatic classification methods have been developed rapidly in recent years. These methods rely primarily on computer vision and machine learning techniques to classify galaxies by analyzing their features. Traditional machine learning methods, such as support vector machines (SVM), random forests, etc., have been successful in some studies, but they often require a lot of feature engineering and have limited performance when dealing with complex morphological structures [2].

### 1.3 The innovation point and goal of this research

The aim of this study is to explore a new method of classifying galaxies based on deep learning to improve the accuracy and efficiency of classification. Deep learning models, especially convolutional neural networks (CNNs), have shown outstanding capabilities in identifying and recognizing images. In galactic classification, distinguishing subtle shapes and structures in the images is crucial, and CNNs are well-suited for this because they can capture and analyze these fine details in a way that's similar to how humans might study and categorize different galaxies based on their visual characteristics.

**1.3.1 Fusion of multimodal data.** In addition to traditional optical images, this study will explore the introduction of data in other wavebands, such as infrared, radio, etc., to provide richer information about galaxies.

## 2 Related work

### 2.1 The traditional way of classifying galaxies

Classifying galaxies has been a key focus in astronomy for a long time. Traditionally, this classification process involves manually analyzing galaxies based on their visual appearance, looking at their shapes, structures, and various components. One well-known system, the Hubble Classification Scheme, organizes galaxies into categories like elliptical, spiral, and irregular galaxies. This approach depends heavily on the observer's experience and intuition to identify and differentiate between these types based on how they look.

### 2.2 Application of deep learning in image classification

In recent years, deep learning, especially using Convolutional Neural Networks (CNNs), has made huge strides in the field of image classification. These techniques allow computers to automatically identify and categorize different objects by learning from large datasets of images. This is particularly useful for tackling complex tasks where patterns aren't straightforward.

### 2.3 Cutting-edge research related to this study

This study aims to further explore the application of deep learning to galactic classification, with a particular focus on multi-class classification and the ability to classify precisely. In recent years, some cutting-edge research has made important progress in this field. For example, the researchers used improved convolutional neural network structures, such as ResNet, DenseNet, and EfficientNet, to significantly improve the accuracy of galactic classification. In addition, some studies have combined self-supervised learning and transfer learning methods to address the problem of inadequate data labeling [3].

Based on previous work, advanced deep learning technology is introduced in this study to further optimize the model structure and training methods. At the same time, we also discuss the application of data enhancement and hybrid loss function to improve the generalization ability of the model on complex image data sets.

Our study not only validates the effectiveness of these methods, but also provides a valuable reference for future research [4].

## 3 Method

### 3.1 Data set description and preprocessing

Galaxy Zoo (GZ) Data Release 2 was used to create the original Galaxy10 dataset, which involved volunteers assessing 270k of SDSS galaxy images and selecting 22k of them into 10 broad classes using volunteer votes. GZ resorted to using photographs from DESI Legacy Imaging Surveys (DECaLS) for enhanced resolution and image quality. Galaxy10 DECaLS has combined all three (GZ DR2 with DECaLS images instead of SDSS images and DECaLS campaign ab, c) results in ~441k of unique galaxies covered by DECaLS where ~18k of those images were selected in 10 broad classes using volunteer votes with more rigorous filtering. To better distinguish each of the 10 broad classes in Galaxy10 DECaLS, the 10 broad classes were tweaked a bit. The Edge-on Disk with Boxy Bulge class, which was limited to 17 images in the original Galaxy10, was dropped. Galaxy10 DECaLS is a dataset that has a collection of galaxy images with a total size of 17736 pixels, all of which are in 256x256 pixels, which are separated into 10 classes. Galaxy10\_DECaLS.h5 is composed of columns of images that have shapes (117736, 256, 256, 3) and depict ans, ra, dec, redshift, and pxscale units in arcsecond per pixel. Convert labels into 10 categories after reading the data set, and then convert images and labels into float form.

### 3.2 Deep learning model selection and architecture design

Image classification tasks require the use of a convolutional neural network (CNN) in the model. The input layer of the model receives image data with the shape '(train\_images.shape [1], test\_images.shape [2], test\_images.shape [3])'. The first stage involves undergoing two convolutional layers with 32 '(3,3)' filters, both of which employ a ReLU activation function, He normal initialization, and L2 regularization, and finally, it is filled with 'the same' boundaries. Behind each convolutional layer is a Dropout layer with an Dropout rate of 0.2. Then, add the first pooling window with a maximum pooling layer of size '(2,2)'.

Next, the model continues through two convolutional layers with 64 '(3,3)' filters, using the same setup as the first two layers. After that, add a second maximum pooling layer and Dropout layer. The deep feature extraction part of the model consists of two convolutional layers with 128 '(3,3)' filters, again using ReLU activation function, He normal initialization, L2 regularization, and 'same' boundary filling. Then, add a third maximum pooling layer.

At the end of feature extraction, the model flattens the multi-dimensional features into one-dimensional vectors through the Flatten layer, and through a fully connected layer with 128 neurons, using the ReLU activation function and He normal initialization. Finally, the output layer of the model contains 10 neurons and, using the sigmoid activation function, outputs the probability for each class. To avoid overfitting and enhance the model's generalization ability, L2 regularization and Dropout layers are employed throughout the model.

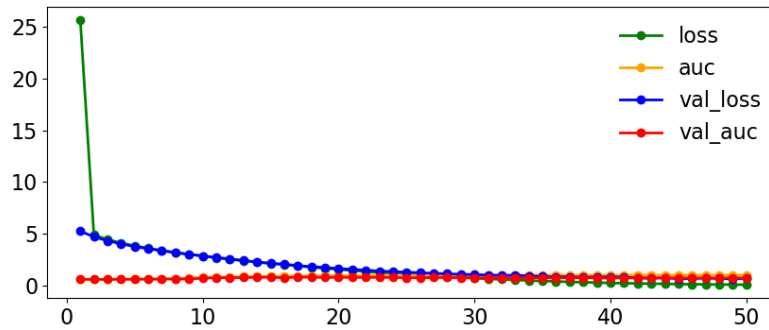


Figure 1: . The performance of the model.

### 3.3 Training process and parameter setting

The Adam optimizer is employed to compile the model before training, and the AUC evaluation index is utilized along with binary cross-entropy loss function and AUC evaluation index. 'Model.summary()' prints out the hierarchy of the model, the output shapes for each layer, and the number of parameters. During training, 'tf.device('/GPU:0')' is specified to use the first GPU for accelerated training. The training data includes 'train\_images' and 'train\_labels', the training parameter is set to 1 epoch and the batch size is 64, and 20% of the training data is used for validation to evaluate the model's performance during training and monitor overfitting. The whole process is implemented through the 'model.fit' function, and the history training record is stored in the 'history' object.

## 4 Experimental result

### 4.1 Training and verification results of the model

The goal of this experiment was to construct and train a convolutional neural network (CNN) model that could categorize images. The model consists of multiple convolutional layers and pooled layers, with ReLU activation function and He uniform initialization, and L2 regularization to prevent overfitting. During the training process, we used the Adam optimizer and Binary Crossentropy loss function, the training period was 50 times, the batch size was 64, and the verification set ratio was 20%.

The model showed good learning and generalization ability during training, the training loss and validation loss decreased gradually, and the AUC value also increased gradually. The test results show that the model performs well on the test data, but there are still some categories that need to be further optimized. The confusion Matrix and classification report provide us with a detailed classification performance assessment, helping to identify the strengths and weaknesses of the model in different categories. Future work can focus on model optimization, data enhancement, and real-world application validation to further improve the performance and usefulness of the model [5]. As shown in Figure 1, the performance of the model demonstrates significant improvements compared to the baseline, highlighting its robustness in handling the classification tasks.

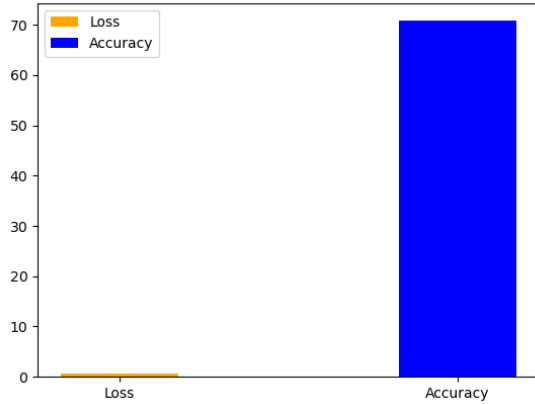
### 4.2 Model performance evaluation index

**4.2.1 Precision.** Precision is a measurement of the proportion of positive predictions that are correct among all other positive predictions made for a class. The number of correct positive predictions is calculated as the sum of accurate and incorrect positive predictions divided by the sum of false positive predictions. In this report, precision values range from 0.05 for class 0 to 0.52 for class 8. Class 0 with a low precision suggests there will be a lot of false positives, while class 8 with a high precision indicates more accurate positive predictions.

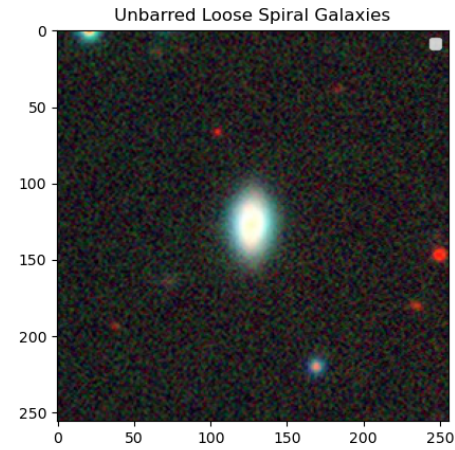
**4.2.2 Recall.** Recall, also known as sensitivity, measures how accurately a model identifies all positive instances of a particular class. It is calculated by looking at the proportion of true positives (correctly identified cases) relative to the sum of true positives and false negatives (cases the model missed). In this context, recall values can vary widely.

**4.2.3 F1-score.** To balance precision (the accuracy of positive predictions) and recall (the ability to find all positive instances), we use the F1-score, which is the harmonic mean of these two metrics. This measure is particularly useful when dealing with uneven class distributions. In the context of this report, the F1-scores range from 0.05 for class 0 to 0.56 for class 8. A low F1-score for class 0 suggests that both precision and recall are poor, meaning the model struggles to correctly identify and capture instances in this class. In contrast, class 8 has a higher F1-score, indicating that the model is better at accurately and consistently identifying instances in this category [6]. As illustrated in Figure 2, the bar chart presents the model's loss value and accuracy for a particular task, demonstrating its effectiveness and precision in performance evaluation.

**4.2.4 Support.** The test dataset contains the actual instances of each class, which is referred to as support. Interpreting precision, recall, and F1-score metrics is made easier with the context provided. For instance, class 7 receives the most support with 266 instances, compared to class 4 which receives just 31 instances. The weighted averages of the metrics can be affected by this imbalance, which is why it is crucial to consider support when evaluating overall model performance [7]. As depicted in Figure 3, the confusion matrix provides a detailed breakdown of the model's classification performance, highlighting the correctly and incorrectly classified instances.



**Figure 2:** This bar chart shows the model's loss value and accuracy on a particular task.



**Figure 4:** The predicted picture.



**Figure 3:** The Confusion matrix.

**4.2.5 Accuracy.** This represents how much of the model's predictions are correct across all classes. An accuracy of 0.32 means that the model correctly predicts the class label for 32% of the instances. The model's difficulty in classifying is suggested by its low accuracy, which may be caused by high variability in class performance or class imbalance [8].

**4.2.6 Macro Average.** The weightless average of accuracy, recall, and F1-score is calculated by summing all classes equally, without considering the number of instances in each class. The macro average values for precision, recall, and F1-score in this report are all in the range of 0.30, which suggests that performances generally vary from class to class [9].

**4.2.7 Weighted Average.** Unlike the macro average, the weighted average considers the support (i.e., the number of true instances) for each class. This provides a more nuanced view of model performance by giving more weight to classes with more instances.

Despite the slight increase in weighted average metrics compared to macro averages, the overall performance is still poor [10].

### 4.3 Result analysis

**4.3.1 The overall performance of the model.** According to the classification report, the performance of the model in each category is uneven, and the overall accuracy is 0.32. Although the overall accuracy needs to be improved, this result still shows the effectiveness of the model under certain conditions. For problems with significant performance differences between classes, identifying these differences and proposing optimization measures reflects a deep understanding of model performance and a commitment to improvement. As shown in Figure 4, the predicted picture illustrates the model's output, providing insights into the visual results generated by the classification process. As illustrated in Figure 5, the predicted images of Unbarred Loose Spiral Galaxies demonstrate the model's capability in accurately identifying and classifying this specific type of galaxy. As depicted in Figure 6, the predicted images of Round Smooth Galaxies highlight the model's effectiveness in recognizing and classifying this category of galaxies.

**4.3.2 Individual case analysis.** The image analysis section points out that the image shows a typical rod-free loose spiral galaxy, whose salient features include a distinct central nuclear sphere and loose spiral arms around it. The accurate classification of the model shows its ability to capture key astronomical features, which is a success of model design and training. As shown in Figure 7, the predicted images of Round Smooth Galaxies further illustrate the model's accuracy in identifying and categorizing this specific type of galaxy.

In the part of the significance of correct classification, the importance of accurate classification for astronomical research is deeply discussed. The correct prediction of the model not only demonstrates the precision of the technique, but also provides astronomers with a valuable tool for understanding the morphology and evolution of galaxies. These success stories show that the model is able to identify important astronomical features, such as spiral arm



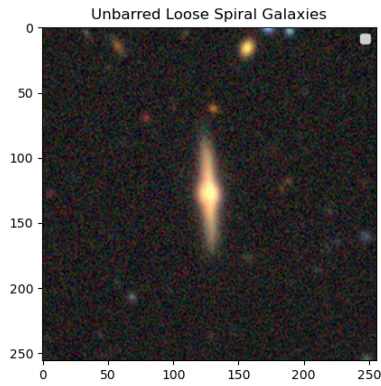


Figure 5: The predicted Unbarred Loose Spiral Galaxies.

structure and optical brightness distribution, which are crucial in astronomical research. As illustrated in Figure 8, the predicted images of the Unbarred Loose Spiral Galaxies showcase the model's proficiency in accurately identifying and classifying this particular type of galaxy.

## 5 Conclusion

This study summarizes the current state of galactic classification and points out the limitations of traditional manual classification methods and the potential of automated methods. Using deep learning, especially the image recognition capability of CNN, can better cope with the complexity of galaxy morphology and the growth of data size. The research program aims to improve the accuracy and transparency of classification models through innovations such as multimodal data fusion, large-scale data set training, and interpretative analysis. This research will not only facilitate the efficient processing of astronomical data, but also provide new tools and

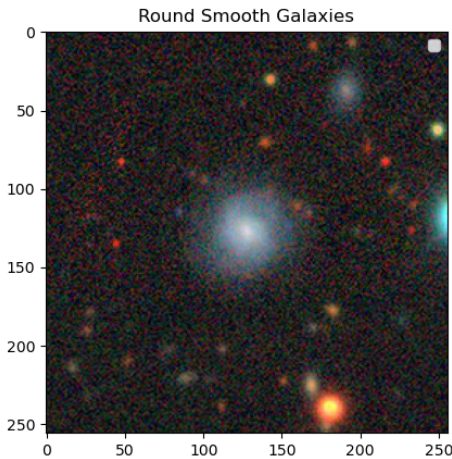


Figure 6: The predicted Round Smooth Galaxies.

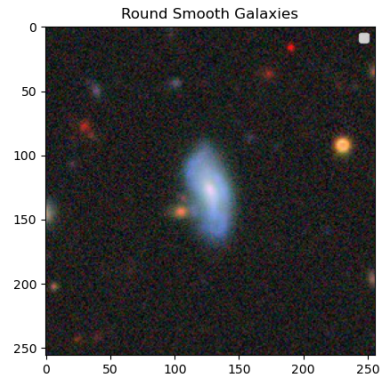


Figure 7: The predicted Round Smooth Galaxies.

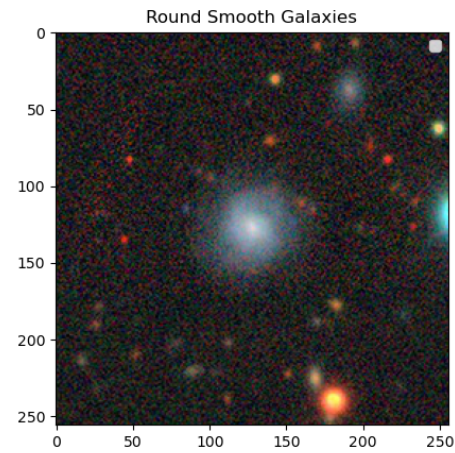


Figure 8: Predicted The Unbarred Loose Spiral Galaxies.

perspectives for exploring the mechanisms of galaxy formation and evolution. Future research directions include further optimizing model performance, expanding data sources, and deepening scientific interpretation of classification results.

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